# Targeted Behavioral Repair with Contaminated Anchors

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## 1 Abstract

We study the problem of correcting a specific, identified behavioral failure in a trained model without retraining from scratch and without full access to the original training distribution. Given a competent base model $\theta_0$ and a small corrective dataset $D_{\text{fix}}$ exhibiting the failure, the standard remedy for avoiding collateral damage elsewhere is to regularize the update toward $\theta_0$ using an anchor set $D_{\text{anc}}$ sampled from the original distribution. We highlight and formalize a complication that this standard recipe does not address: the anchor set is not guaranteed to be failure-free. When the failure region has nonzero mass under the original distribution, $D_{\text{anc}}$ may itself contain instances of the very behavior being corrected, so a preservation term applied indiscriminately can directly oppose the corrective signal. We combine parameter-space preservation (Elastic Weight Consolidation) with output-space preservation (Knowledge Distillation) into a single objective, characterize the repair–preservation trade-off this creates as a function of failure prevalence $\rho$ and dataset sizes, and analyze the resulting multi-step dynamics locally. The analysis shows that, under standard local assumptions, repeated updates converge to a curvature-weighted repair displacement determined by the preservation terms, while anchor contamination perturbs that equilibrium continuously, with the empirical estimate indicating a non-negligible contamination rate. We describe a concrete instantiation used to repair a behavioral-cloning policy trained on the StrikeForce environment.

## 2 Introduction

### 2.1 Problem

A model $\theta_0$ that is already competent on its original task can still exhibit a specific, identifiable failure mode on a subset of the input space. Retraining from scratch is often infeasible: the original training set may no longer be fully available, or reacquiring it is costly relative to simply collecting examples of the failure itself.

### 2.2 Why Naive Fine-Tuning Fails

Fine-tuning $\theta_0$ directly on failure examples $D_{\text{fix}}$ corrects the targeted behavior but risks catastrophic forgetting [1]: nothing in the objective discourages the update from disrupting behavior the model already had right. The standard fix is to add a preservation term computed on an anchor set drawn from the original distribution, whether at the parameter level (EWC) or the output level (KD) [1, 2, 3].

### 2.3 The Contaminated-Anchor Complication

This standard recipe implicitly assumes the anchor set represents only desirable behavior. That assumption is not free: if the failure region $\mathcal{F}$ has nonzero probability $\rho = P_{\text{orig}}(x \in \mathcal{F})$ under the original distribution, an anchor set sampled from that same distribution will, in expectation, contain a $\rho$-fraction of failure examples. On those examples, the preservation term explicitly pulls the model back toward the very behavior the corrective term is trying to remove. We refer to this as the **contaminated-anchor** setting and treat it as a first-class part of the problem rather than an edge case to be assumed away.

### 2.4 Our Approach

We combine EWC (parameter-space preservation, weighted by Fisher-style sensitivity) and KD (output-space preservation through temperature-scaled softmax agreement with $\theta_0$) into a single joint objective (Section 4). We then analyze the resulting multi-step repair dynamics and the effect of anchor contamination on the local preservation geometry (Section 6), followed by the practical repair–preservation trade-off in terms of $\rho$, $|D_{\text{fix}}|$, $|D_{\text{anc}}|$, and the regularization strengths $\lambda_1, \lambda_2$ (Section 7). We do not claim the contamination problem is solved outright; rather, we make its effect explicit so that the method's behavior under it is predictable rather than silently absorbed into noise.

### 2.5 Summary of Contributions

- A formalization of the contaminated-anchor setting for behavioral repair, distinguishing it from standard continual-learning preservation, which typically assumes clean task boundaries.
- A combined EWC+KD objective (Section 4) with an explicit account of how the corrective and preservation gradients interact on contaminated anchor points (Section 5).
- A local multi-step analysis of the repair dynamics, showing that the joint update converges, under standard local assumptions, to a curvature-weighted repair displacement determined by the preservation geometry (Section 6).
- A characterization of how anchor contamination perturbs that preservation geometry and therefore the resulting repair point as a function of failure prevalence $\rho$ (Section 6).
- A concrete implementation and repair pipeline applied to a behavioral-cloning policy for the StrikeForce environment (Section 8), with pre-repair ($\theta_0$), post-repair ($\theta^\star$), and naive fine-tuning ($\theta_{\text{FT}}$) checkpoints evaluated over $n = 64$ paired episodes on base, derived, and composite engagement metrics (Section 11).

# 3 Problem Setting

Let $f_\theta : \mathcal{X} \to \mathbb{R}^K$ be a trained model with parameters $\theta$. We assume an initial model $\theta_0$ is already competent on the original task distribution $P_{\text{orig}}$, but exhibits a specific behavioral failure on a subset $\mathcal{F} \subseteq \mathcal{X}$. The goal is not to retrain from scratch, but to modify $\theta_0$ so that the undesired behavior is corrected while preserving the model's behavior outside $\mathcal{F}$.

We assume access to:

- a **corrective dataset** $D_{\text{fix}} = \{(x_i, y_i)\}_{i=1}^{n_f}$, containing examples of the undesired behavior with corrected labels;
- an **anchor dataset** $D_{\text{anc}} = \{x_j\}_{j=1}^{n_a}$, sampled from $P_{\text{orig}}$;
- the original parameters $\theta_0$ and the original model's outputs on the anchor examples.

The original training set itself is assumed unavailable or too costly to reacquire, a setting also studied in continual-learning work such as Learning without Forgetting [3]. The operational criterion used to construct $D_{\text{fix}}$ is described in Section 8.3.

**Definition 1** (Contaminated-Anchor Assumption)**.** *Let $\rho = P_{\text{orig}}(x \in \mathcal{F}) > 0$ denote the failure prevalence under the original distribution. Since $D_{\text{anc}}$ is sampled from $P_{\text{orig}}$, $\rho$ is equivalently the expected failure frequency* within *$D_{\text{anc}}$; it is* not *the failure frequency within the corrective set $D_{\text{fix}}$, which is failure-dominated by construction: every frame in $D_{\text{fix}}$ is an expert action recorded during an intervention.*

$$P_{\text{orig}} = (1 - \rho)\, P_{\text{good}} + \rho\, P_{\text{fail}},$$

*and is therefore* not *assumed failure-free: $D_{\text{anc}} \cap \mathcal{F} \neq \varnothing$ in general.*

This creates the central tension of the problem: a preservation mechanism applied indiscriminately to all anchor samples can oppose the desired repair.

# 4 Combined Objective

We seek $\theta^\star \approx \theta_0$ in the sense of preserving general behavior, while improving performance on $\mathcal{F}$. The update $\Delta\theta = \theta^\star - \theta_0$ should be targeted rather than globally disruptive.

## 4.1 Corrective Term

$$L_{\text{fix}}(\theta) = \frac{1}{n_f} \sum_{(x_i, y_i) \in D_{\text{fix}}} \ell(f_\theta(x_i), y_i).$$

$D_{\text{fix}}$ is not intended to represent the full original distribution; it supplies a targeted signal for the specific missing or incorrect behavior.

**Choice of $\ell$ and the detached focal weight.** We use a focal-style per-example loss

$$\ell(f_\theta(x), y) = -\big(1 - p_t(x;\theta)\big)^\gamma \log p_t(x;\theta),$$

where $p_t(x;\theta)$ is the model's softmax probability on the target class and $\gamma \geq 0$ is the focal parameter. Throughout the analysis we treat the modulating factor $w(x;\theta) = (1 - p_t(x;\theta))^\gamma$ as a *detached* per-example weight: it is evaluated at the current parameters and is treated as a constant with respect to differentiation. This is a standard implementation choice for focal-style losses and is used here so

that the per-example loss reduces to a weighted cross-entropy with a parameter-dependent weight. Under this convention the per-example gradient with respect to the model logit is

$$\frac{\partial \ell}{\partial z} = w(x;\theta)\,\big(p_t(x;\theta) - 1\big),$$

and the corresponding per-example second derivative is

$$\frac{\partial^2 \ell}{\partial z^2} = w(x;\theta)\,p_t(x;\theta)\big(1 - p_t(x;\theta)\big) \geq 0.$$

This non-negativity is what makes the corrective term locally convex in the model's own output, a property we use in Section 6.

### 4.2 Parameter-Space Preservation (EWC)

For each anchor example $x_j$, let $\ell_{\text{anc}}(x_j;\theta_0)$ denote the per-example loss used to characterize the original model's behavior on that example. In the current implementation, the diagonal Fisher-style coefficient for parameter $\theta_i$ is estimated by accumulating the squared per-sample gradients and averaging over the anchor set:

$$\widehat{F}_i = \frac{1}{n_a}\sum_{j=1}^{n_a}\left(\frac{\partial \ell_{\text{anc}}(x_j;\theta_0)}{\partial \theta_i}\right)^2.$$

Thus $\widehat{F}_i$ is the mean squared sensitivity of parameter $\theta_i$ to the anchor-set loss. When the per-example loss is a negative log-likelihood under the usual Fisher construction, this is an empirical diagonal Fisher approximation; more generally, it should be interpreted as a squared-gradient sensitivity statistic. We use $\widehat{F}_i$ in the EWC penalty

$$L_{\text{EWC}}(\theta) = \frac{1}{2}\sum_i \widehat{F}_i\,(\theta_i - \theta_{0,i})^2\,.$$

Parameters with larger $\widehat{F}_i$ therefore incur a larger cost for moving away from $\theta_{0,i}$ [1].

### 4.3 Output-Space Preservation (KD)

For anchor input $x_j$, let $z_0(x_j) = f_{\theta_0}(x_j)$ and $z_\theta(x_j) = f_\theta(x_j)$. With temperature $T$,

$$p_0^T(x_j) = \text{softmax}\left(\frac{z_0(x_j)}{T}\right), \qquad p_\theta^T(x_j) = \text{softmax}\left(\frac{z_\theta(x_j)}{T}\right),$$

and we use a forward-KL output-preservation objective,

$$L_{\text{KD}}(\theta) = \frac{1}{n_a}\sum_{j=1}^{n_a}\text{KL}\big(p_\theta^T(x_j)\,\|\,p_0^T(x_j)\big) = \frac{1}{n_a}\sum_{j=1}^{n_a}\sum_{k=1}^{K} p_{\theta,k}^T(x_j)\log\frac{p_{\theta,k}^T(x_j)}{p_{0,k}^T(x_j)}.$$

This is a KL-based behavioral anchoring mechanism related to the distillation literature [2, 4]; the present formulation specifically uses the positive forward KL implemented by the repair procedure.

### 4.4 Combined Objective

$$L_{\text{total}}(\theta) = L_{\text{fix}}(\theta) + \lambda_1 L_{\text{EWC}}(\theta) + \lambda_2 L_{\text{KD}}(\theta), \qquad \lambda_1, \lambda_2 \geq 0.$$

# 5 The Conflict Induced by Anchor Contamination

If every anchor example represented desirable behavior, $\nabla_\theta L_{\text{EWC}}$ and $\nabla_\theta L_{\text{KD}}$ would generally align with the goal of behavioral preservation without opposing repair. Under Definition 1, this no longer holds unconditionally.

For $x \in D_{\text{anc}} \cap \mathcal{F}$, the reference output $f_{\theta_0}(x)$ is precisely the behavior the corrective term seeks to change. The KD gradient at such a point pulls $p_\theta^T(x)$ toward $p_0^T(x)$, i.e., back toward the failure behavior, directly opposing

$$\nabla_\theta L_{\text{fix}} \quad \text{vs.} \quad \nabla_\theta L_{\text{KD}}.$$

The method does not attempt to preserve $\theta_0$'s behavior everywhere; it seeks an update in which the corrective signal is strong enough to overcome preservation pressure specifically on $\mathcal{F}$, while preservation remains effective on the (typically much larger) non-failure portion of the anchor set.

**Remark 1.** *This is an asymmetry, not a paradox: $D_{\text{fix}}$ is concentrated on $\mathcal{F}$ by construction, while the contaminated fraction of $D_{\text{anc}}$ is only $\rho \cdot n_a$ examples in expectation. The repair signal on $\mathcal{F}$ can dominate provided $|D_{\text{fix}}|$ and its effective weighting are large relative to $\lambda_2 \rho\, n_a$. Because the two gradient fields are architecture-, data-, and loss-dependent, we do not claim a closed-form sufficient condition for dominance; the present paper treats this dependence as a qualitative trade-off to be evaluated empirically rather than as a universal analytical guarantee.*

# 6 Local Multi-Step Repair Dynamics

The key theoretical question is not whether the corrective gradient dominates on the first optimization step, but where the sequence of updates moves after many steps. Define the preservation objective

$$R(\theta) = \lambda_1 L_{\text{EWC}}(\theta) + \lambda_2 L_{\text{KD}}(\theta),$$

so that

$$L_{\text{total}}(\theta) = L_{\text{fix}}(\theta) + R(\theta).$$

The repair procedure follows

$$\theta_{t+1} = \theta_t - \eta \left[\nabla L_{\text{fix}}(\theta_t) + \nabla R(\theta_t)\right].$$

## 6.1 Local Curvature of the Corrective and Preservation Terms

We work with the second-order expansion of $L_{\text{total}}$ around $\theta_0$. Let

$$H_F = \nabla^2 L_{\text{fix}}(\theta_0), \qquad H_R = \nabla^2 R(\theta_0), \qquad H_{\text{total}} = H_F + H_R,$$

denote the corrective, preservation, and total Hessians at the original parameters, and let

$$g_F = \nabla L_{\text{fix}}(\theta_0).$$

Because both preservation terms are minimized at $\theta_0$, we have $R(\theta_0) = 0$ and $\nabla R(\theta_0) = 0$, so $g_F$ is the only first-order term in the expansion.

**Corrective curvature.** By Section 4.1, the detached focal weight makes the per-example second derivative in the target logit non-negative. For a multi-class softmax output $p(x;\theta) \in \Delta^{K-1}$, the per-example Hessian of $\tilde{L}_{\text{fix}}$ with respect to the logits is

$$C_F(x) = w(x;\theta_0)\left[\operatorname{Diag}\left(p(x;\theta_0)\right) - p(x;\theta_0)\,p(x;\theta_0)^\top\right] \succeq 0,$$

which is the standard softmax Fisher matrix scaled by the detached focal weight $w(x;\theta_0) \geq 0$. Writing $J_F(x)$ for the Jacobian of the logits with respect to $\theta$ at $\theta_0$, the parameter-space Hessian of the corrective term is

$$H_F = \frac{1}{n_f}\sum_{i=1}^{n_f} J_F(x_i)^\top C_F(x_i) J_F(x_i) \succeq 0.$$

Thus, under the detached-focal convention, the corrective term is locally convex in the parameters up to the rank structure of the Jacobians.

**Preservation curvature.** For the EWC term,

$$H_{\text{EWC}} = \operatorname{diag}(\widehat{F}_1, \ldots, \widehat{F}_d),$$

so the mean-squared per-sample gradient estimate enters the local preservation geometry directly as diagonal curvature.

For the KL term, let $J_j$ be the Jacobian of the model logits with respect to $\theta$ at $\theta_0$ for anchor $x_j$, and define

$$C_j^T = \frac{1}{T^2}\left[\operatorname{Diag}(p_0^T(x_j)) - p_0^T(x_j)\,p_0^T(x_j)^\top\right].$$

At $\theta_0$, the Hessian of the KL term with respect to parameters is

$$H_{\text{KD}} = \frac{1}{n_a}\sum_{j=1}^{n_a} J_j^\top C_j^T J_j \succeq 0.$$

Therefore,

$$H_R = \lambda_1 \operatorname{diag}(\widehat{F}) + \lambda_2 H_{\text{KD}} \succeq 0.$$

**Total curvature.** Combining the two, locally at $\theta_0$,

$$\boxed{H_{\text{total}} = H_F + \lambda_1 \operatorname{diag}(\widehat{F}) + \lambda_2 H_{\text{KD}} + \gamma I \succ 0}$$

where $\gamma > 0$ is a small fixed damping constant. Absorbing $\gamma I$ into the definition of $H_{\text{total}}$ ensures positive definiteness on the full parameter space and makes the equilibrium $\Delta^\star = -H_{\text{total}}^{-1} g_F$ well defined without a separate case distinction.

The corrective term contributes curvature that locally at $\theta_0$. reflects where the model's own output is still uncertain on the corrective set, the EWC component penalizes parameter-space displacement according to squared gradient sensitivity, and the KD component penalizes displacement that changes the model outputs on the anchors. We assume throughout the analysis that, with the damping $\gamma I$ included, $H_{\text{total}}$ has eigenvalues in $[\mu, L]$ on the subspace $\mathcal{S}$ spanned by the optimization trajectory, for some $0 < \mu \leq L$. Since $\gamma > 0$, the lower bound $\mu \geq \gamma$ holds automatically and the equilibrium below is well defined.

### 6.2 Multi-Step Dynamics

Within a sufficiently small neighborhood of $\theta_0$, retaining the quadratic terms of both the corrective and preservation objectives gives the local surrogate

$$L_{\text{local}}(\Delta) = L_{\text{total}}(\theta_0) + g_F^\top \Delta + \frac{1}{2}\Delta^\top H_{\text{total}} \Delta, \qquad \Delta = \theta - \theta_0.$$

The corresponding multi-step dynamics are

$$\Delta_{t+1} = (I - \eta H_{\text{total}})\Delta_t - \eta g_F.$$

The corrective gradient $g_F$ drives the model away from $\theta_0$, while the total curvature $H_{\text{total}}$ grows the restoring force with displacement. Both the preservation terms and the local curvature of the corrective term contribute to this restoring effect; the focal-detached weight $w$ sets the scale of the corrective contribution.

**Proposition 1** (Local Equilibrium of the Repair Dynamics)**.** *With $H_{\text{total}}$ defined as in Section 6.1 (including the damping $\gamma I$), let its eigenvalues on the subspace $\mathcal{S}$ containing the optimization trajectory lie in $[\mu, L]$, where $0 < \mu \leq L$, and suppose the local surrogate above is valid. If*

$$0 < \eta < \frac{2}{L},$$

*then gradient descent on the local surrogate converges linearly to the unique stationary point*

$$\boxed{\Delta^\star = -H_{\text{total}}^{-1} g_F}$$

*on $\mathcal{S}$. Moreover,*

$$\|\Delta_t - \Delta^\star\| \leq q^t \|\Delta_0 - \Delta^\star\|, \qquad q = \max_{h \in [\mu, L]} |1 - \eta h| < 1.$$

*Proof.* The stationary-point equation of the local surrogate is

$$g_F + H_{\text{total}} \Delta^\star = 0,$$

which yields $\Delta^\star = -H_{\text{total}}^{-1} g_F$. Subtracting the fixed-point equation from the update gives

$$\Delta_{t+1} - \Delta^\star = (I - \eta H_{\text{total}})(\Delta_t - \Delta^\star).$$

Because the eigenvalues of $H_{\text{total}}$ lie in $[\mu, L]$ and $0 < \eta < 2/L$, the spectral radius of $I - \eta H_{\text{total}}$ is strictly smaller than one, giving the stated linear convergence bound. □

This result formalizes the multi-step interpretation of the method. The repair does not simply follow the corrective gradient indefinitely. Instead, the corrective signal drives the model away from $\theta_0$, while the total curvature—comprising both the corrective curvature and the preservation curvature—increasingly resists displacement. Training settles when these forces balance:

$$\boxed{\nabla L_{\text{fix}}(\theta^\star) + \nabla R(\theta^\star) = 0.}$$

Under the local model, this balance is precisely $\Delta^\star = -H_{\text{total}}^{-1} g_F$.

### 6.3 Interpretation as a Curvature-Weighted Repair

The total Hessian $H_{\text{total}}$ is symmetric positive definite, so it admits the spectral decomposition

$$H_{\text{total}} = Q\Lambda Q^\top, \qquad \Lambda = \text{diag}(h_1, \ldots, h_d),$$

where the columns $v_1, \ldots, v_d$ of $Q$ are an orthonormal basis of eigenvectors and $h_i \geq 0$ are the associated total curvatures. With the damping $\gamma I$ included in $H_{\text{total}}$, we have $h_i \geq \gamma > 0$ for all $i$.

Working in the eigenbasis of $H_{\text{total}}$, let

$$\tilde{\Delta} = Q^\top \Delta, \qquad \tilde{g}_F = Q^\top g_F,$$

so that $\tilde{\Delta}_i = \langle \Delta, v_i \rangle$ is the component of the parameter displacement along the $i$-th eigenvector of the total Hessian. The stationary condition $\tilde{g}_F + \Lambda \tilde{\Delta}^\star = 0$ then decouples coordinate-wise,

$$\tilde{\Delta}_i^\star = -\frac{(\tilde{g}_F)_i}{h_i}.$$

The equilibrium displacement therefore permits larger movement along eigenvectors with small total curvature $h_i$ and suppresses movement along eigenvectors with large $h_i$. The resulting repair direction is not the raw corrective gradient direction; it is the corrective gradient filtered through the spectrum of the total curvature, i.e. through what the corrective term, the EWC term, and the KD term jointly consider important. In particular, the corrective curvature $H_F$ can only *increase* the $h_i$ of directions in which the model is uncertain on $D_{\text{fix}}$, thereby further limiting the equilibrium displacement along those directions.

The same spectral picture controls both the achievable repair gain and the preservation cost. Under the local surrogate,

$$L_{\text{fix}}(\theta_0 + \Delta^\star) - L_{\text{fix}}(\theta_0) \approx -g_F^\top H_{\text{total}}^{-1} g_F = -\sum_{i=1}^{d} \frac{(\tilde{g}_F)_i^2}{h_i},$$

whereas

$$R(\theta_0 + \Delta^\star) \approx \frac{1}{2} g_F^\top H_{\text{total}}^{-1} H_R H_{\text{total}}^{-1} g_F.$$

If $H_R$ and $H_F$ are simultaneously diagonalized by the same basis (a simplification that holds exactly only when they commute), the preservation cost reduces to

$$R(\theta_0 + \Delta^\star) \approx \frac{1}{2} \sum_{i=1}^{d} \frac{(\tilde{g}_F)_i^2}{h_i} \cdot \frac{(h_R)_i}{h_i},$$

so that the ratio $(h_R)_i / h_i$ controls how much of the corrective displacement is "paid for" in preservation cost along each direction. The quantity

$$S \;=\; g_F^\top H_{\text{total}}^{-1} g_F$$

therefore summarizes the achievable first-order repair gain under the local quadratic model, with the preservation cost bounded by $S/2$ in the commuting case and by $\frac{1}{2} h_{\max}(H_R) \, \| H_{\text{total}}^{-1} g_F \|^2$ in general. The value of $S$ is larger when the corrective gradient has substantial components along eigenvectors of low total curvature, which is precisely the regime in which the repair can be achieved with limited collateral movement. Conversely, when the corrective gradient is concentrated along high-curvature eigenvectors, $S$ is small: the combined curvature suppresses the displacement and the achievable repair gain is correspondingly limited.

### 6.4 Behavior-Level Targetedness

Parameter displacement alone does not imply behavioral locality. To connect the parameter-space result to the desired behavioral property, let $z_G(\theta)$ and $z_F(\theta)$ denote the stacked model outputs on representative good and failure inputs, respectively, with Jacobians at $\theta_0$

$$J_G = \left.\frac{\partial z_G}{\partial \theta}\right|_{\theta_0}, \qquad J_F = \left.\frac{\partial z_F}{\partial \theta}\right|_{\theta_0}.$$

At the local equilibrium,

$$z_G(\theta^\star) - z_G(\theta_0) \approx -J_G H_{\text{total}}^{-1} g_F,$$

while

$$z_F(\theta^\star) - z_F(\theta_0) \approx -J_F H_{\text{total}}^{-1} g_F.$$

Therefore, a sufficient local condition for a targeted repair is that the total-curvature-weighted corrective direction has a substantially larger effect on the failure outputs than on the good outputs, for example

$$\boxed{\|J_F H_{\text{total}}^{-1} g_F\| \gg \|J_G H_{\text{total}}^{-1} g_F\|.}$$

A stronger structural version is the existence of a repair direction $d$ such that

$$g_F^\top d < 0, \qquad \|J_G d\| \approx 0, \qquad \|J_F d\| > 0.$$

Such a direction improves the corrective objective while producing little first-order output change on good behavior. The combined EWC+KD+corrective geometry favors precisely those parameter directions that can produce the required repair with lower total curvature. This does not constitute a global guarantee of behavioral equivalence; it is a local sufficient condition linking the parameter-space equilibrium to the desired behavior-level locality.

### 6.5 Effect of Anchor Contamination

Under the contaminated-anchor model,

$$P_{\text{anc}} = (1-\rho)P_{\text{good}} + \rho P_{\text{fail}},$$

the population version of the preservation Hessian decomposes accordingly:

$$H_R(\rho) = (1-\rho)H_{R,\text{good}} + \rho H_{R,\text{fail}} + H_{\text{shared}},$$

where $H_{\text{shared}}$ denotes any $\rho$-independent contribution such as a separately fixed preservation component. Equivalently, after absorbing the common term into $H_R(0)$,

$$H_R(\rho) = H_R(0) + \rho\,(H_{R,\text{fail}} - H_{R,\text{good}})\,.$$

The corrective Hessian $H_F$ does not depend on $\rho$, since $D_{\text{fix}}$ is failure-dominated by construction and is not resampled from $P_{\text{orig}}$. Thus the total Hessian inherits the contamination dependence through $H_R$ alone:

$$H_{\text{total}}(\rho) = H_F + H_R(\rho) = H_{\text{total}}(0) + \rho\,(H_{R,\text{fail}} - H_{R,\text{good}})\,.$$

Let

$$E_\rho = \rho\,(H_{R,\text{fail}} - H_{R,\text{good}})\,.$$

If $H_{\text{total}}(0) \succeq \mu I$ and

$$\|E_\rho\| < \mu,$$

then $H_{\text{total}}(\rho)$ remains positive definite and standard matrix perturbation gives

$$\|H_{\text{total}}(\rho)^{-1} - H_{\text{total}}(0)^{-1}\| \leq \frac{\|E_\rho\|}{\mu\,(\mu - \|E_\rho\|)}.$$

Consequently, with the same local corrective gradient $g_F$,

$$\boxed{\|\Delta^\star(\rho) - \Delta^\star(0)\| \leq \frac{\|E_\rho\|}{\mu\,(\mu - \|E_\rho\|)}\|g_F\| = O(\rho)}$$

for sufficiently small $\rho$. Hence rare failure cases in the anchor set perturb the local repair equilibrium continuously rather than changing it discontinuously. The contamination remains harmful because its curvature points toward preserving the old failure behavior, but when $\rho$ is small its aggregate influence is correspondingly limited under the stated conditioning assumptions. When $\ker(H_R) \subseteq \text{range}(H_F)$, the corrective curvature $H_F$ raises the lower bound $\mu$ in the perturbation bound and reduces the sensitivity of the equilibrium to anchor contamination. Outside this case, $H_F$ does not change $\mu$ and the bound is governed by the preservation geometry alone.

The bound is stated under the perturbative condition $\|E_\rho\| < \mu$ and is a first-order statement in $\rho$; it is not claimed to hold at the empirically estimated $\hat{\rho} \approx 0.185$ (Section 8.3), where the condition is correspondingly harder to satisfy. The result should therefore be read as characterizing the local sensitivity of the repair equilibrium to contamination in the small-$\rho$ regime, not as a guarantee that the shift is negligible at the operating point of the experiments.

### 6.6 Theoretical Interpretation

The resulting picture is therefore a multi-step balance rather than a one-step dominance argument:

$$\underbrace{\nabla L_{\text{fix}}}_{\text{repair drive}} + \underbrace{\lambda_1 \nabla L_{\text{EWC}} + \lambda_2 \nabla L_{\text{KD}}}_{\text{restoring preservation force}} = 0 \quad \text{at equilibrium,}$$

and the local geometry of this balance is governed by the total Hessian

$$H_{\text{total}} = H_F + \lambda_1 \,\text{diag}(\widehat{F}) + \lambda_2 H_{\text{KD}}.$$

The corrective curvature $H_F$ reflects where the model is still uncertain on the corrective set; the EWC coefficients measured as mean squared per-sample sensitivities determine which parameter directions are expensive to change; KD determines which anchor-output changes are expensive. When the corrective signal admits a low-total-curvature direction, the equilibrium can produce a substantial correction with limited collateral movement. Anchor contamination perturbs $H_R$ and therefore $H_{\text{total}}$ in proportion to its prevalence, so the main failure mode of contamination is a shift of the equilibrium toward preserving part of the behavior that the repair objective seeks to eliminate.

The analysis is local and does not claim that neural-network optimization is globally convex, that the empirical Fisher equals the exact Fisher in every implementation, or that the resulting equilibrium necessarily gives perfect failure correction and zero general-performance loss.

# 7 Repair–Preservation Trade-off

As $\rho$ decreases, most anchor samples represent desirable behavior and the preservation terms receive a strong aggregate signal from non-failure examples, reducing the conflict of Section 5. In the local analysis of Section 6, this corresponds to a smaller perturbation of the preservation Hessian $H_R(\rho)$. The corrective dataset provides a concentrated signal $D_{\text{fix}} \sim P(x \mid x \in \mathcal{F})$ regardless of $\rho$. The effective optimization is therefore governed jointly by $|D_{\text{fix}}|$, $|D_{\text{anc}}|\,\rho$, the preservation geometry, and the regularization strengths $\lambda_1, \lambda_2$, rather than by either dataset size alone.

We do not assume a fixed value of $\lambda_1, \lambda_2$ is universally optimal; the method should be evaluated as a function of $|D_{\text{fix}}|, |D_{\text{anc}}|, \rho, \lambda_1, \lambda_2$.

### Target Property

The desired solution satisfies, qualitatively,

$$\text{Behavior}_{\text{fail}}(\theta^\star) \succ \text{Behavior}_{\text{fail}}(\theta_0), \qquad \text{Performance}_{\text{good}}(\theta^\star) \approx \text{Performance}_{\text{good}}(\theta_0),$$

where $\text{Behavior}_{\text{fail}}(\theta)$ denotes the operational failure behavior of $\theta$ on the intervention-defined failure region of Section 8.3, and $\succ$ means strictly better on that axis. The objective is a *localized* behavioral correction with limited collateral change, not merely a lower training loss.

# 8 Implementation

We instantiate the combined objective for a behavioral-cloning policy [9] trained in the *StrikeForce* 2D partially observable battle-arena testbed [10], using the decoupled Future Action Prediction (FAP)/Transformer-aggregator architecture described in [11]. We therefore apply the repair directly to the converged checkpoint $\theta_0$ produced by that training procedure, without independently retraining the underlying policy architecture for the present study.

## 8.1 Precomputed Anchor Statistics

Before repair, we precompute and pin, for each anchor episode: $\theta_0$ itself, an estimate of the Fisher diagonal $\{F_i\}$, and the reference future-prediction and policy logits $\{z_0(x_j)\}$ under $\theta_0$. Pinning the logits rather than recomputing $f_{\theta_0}(x_j)$ at every step avoids carrying a second live copy of the model during repair and fixes the KD reference exactly.

## 8.2 Per-Step Composition

At each optimization step we compute three gradient contributions and sum them before the optimizer step: (i) the corrective gradient from a shuffled minibatch of $D_{\text{fix}}$, using the same time-discounted focal loss as the original training procedure; (ii) the EWC gradient, evaluated over all parameters against the pinned $(\theta_0, F)$; (iii) the KD gradient, evaluated as a KL-type divergence between current and pinned softmax outputs at both the future-prediction heads and the main policy head.

**Remark 2.** *In the current implementation, the corrective term is minibatched (shuffled subsets of $D_{\text{fix}}$ each step) while the KD term is evaluated over the entire anchor set at every step. This gives the anchor term zero gradient stochasticity relative to the corrective term and dominates per-step compute. The asymmetry is intentional: the anchor term is kept deterministic across*

*optimization steps so that the preservation signal is applied consistently against the fixed reference outputs. This does not change the joint-objective gradient identity of Section 4, and full-batch KD is an implementation choice rather than a requirement of the method; minibatched KD would be a valid computational alternative.*

### 8.3 Anchor and Corrective Set Provenance

The corrective set $D_{\text{fix}}$ is constructed from *intervention episodes* collected during unmasked inference. Concretely, while the policy is deployed in inference mode, it occasionally enters a state in which it produces an unpleasant, illogical, or otherwise unacceptable action. A human expert intervenes at that point, takes control, and executes the intended action until the policy has escaped the problematic situation, after which inference resumes. Each intervention episode therefore supplies a short trajectory of states paired with the expert's actions, i.e. the corrected labels for the failure region. During repair, the model is trained only on these expert actions. The corrective set is thus concentrated on $\mathcal{F}$ by construction, and its failure density is close to one, in contrast to the anchor set. Operationally, the failure region $\mathcal{F}$ in this study corresponds to states in which the policy disengages from combat when engagement is warranted; the expert interventions recorded in $D_{\text{fix}}$ are precisely the actions that restore engagement. The composite metrics of Section 11.3 are designed to measure this axis directly.

The anchor set $D_{\text{anc}}$, by contrast, is sampled from the original distribution via the stage-1 validation episodes. It is not constructed to exclude failure cases: the behavioral failure is identifiable but is not assumed to be cleanly separable from the original distribution. Instead, the intended setting is that such cases are relatively rare, so most anchors still represent desirable behavior while a minority may overlap with $\mathcal{F}$. This is precisely the contamination scenario formalized in Definition 1.

Here $\rho$ denotes the empirical failure frequency *within* $D_{\text{anc}}$, not within $D_{\text{fix}}$: the two sets have very different failure densities by construction. Every frame in $D_{\text{fix}}$ is an expert action recorded during an intervention, so the failure density in $D_{\text{fix}}$ is close to one. The anchor set, by contrast, contains only the $\rho$-fraction of frames that coincide with the failure region.

We estimate $\rho$ empirically from the rate at which the policy required expert intervention during normal operation, using the intervention episodes that also supply $D_{\text{fix}}$; this rate serves as a proxy for $P_{\text{orig}}(x \in \mathcal{F})$ under the assumption that the deployment episodes are representative of the original distribution. Across 28 episodes of 1024 frames each (28,672 frames total), 5300 frames required intervention, giving

$$\hat{\rho} = \frac{5300}{28672} \approx 0.185.$$

This estimate is non-negligible: the contaminated fraction of $D_{\text{anc}}$ is a substantial minority rather than a vanishing edge case. The contaminated-anchor setting formalized in this paper is therefore a practical concern rather than a theoretical curiosity, and the conflict between the corrective and preservation gradients analyzed in Section 5 is expected to be observable in the experiments rather than masked by a vanishing $\rho$. When episode-level labels or behavioral inspection are available, the observed failure frequency in $D_{\text{anc}}$ can be reported directly as an empirical estimate of the contamination rate; the method itself does not require explicit filtering of those examples.

### 8.4 Hyperparameters

We use $T = 1$ in the current implementation, i.e., ordinary softmax rather than temperature-softened distillation targets. The regularization strengths of the combined objective are set to

$$\lambda_1 = \lambda_2 = 5,$$

i.e., the EWC and KD preservation terms are weighted equally in Section 4. This is the configuration used for the $\theta^\star$ checkpoint reported in Section 11; the values were fixed for the repair runs and were not separately tuned per metric or per failure mode. As stated in Section 7, the trade-off between repair and preservation is governed jointly by $(|D_{\text{fix}}|, |D_{\text{anc}}|, \rho, \lambda_1, \lambda_2)$, so this choice is a single operating point on that trade-off rather than a claim of optimality. The naive fine-tuning baseline $\theta_{\text{FT}}$ corresponds to the same pipeline with $\lambda_1 = \lambda_2 = 0$.

## 9 Related Work

Elastic Weight Consolidation [1] penalizes movement of parameters deemed important to previous tasks, estimated via the Fisher information matrix; we use the same diagonal-Fisher penalty but apply it against a single identified failure region rather than a sequence of task boundaries. Knowledge distillation [2] was originally proposed for transferring information from a teacher to a student; here $\theta_0$ and $\theta^\star$ share an architecture, and KD instead serves as an output-space behavioral anchor. Learning without Forgetting [3] is also closely related to our data-availability constraint because it studies learning new capabilities when the original training data are unavailable, although its objective is preservation of existing capabilities while adding new ones rather than correcting a known failure.

More broadly, our setting belongs to the family of post-hoc model-editing approaches that seek to alter a pretrained model while limiting unintended changes. MEND [5] learns to transform gradients for fast, localized edits, ROME [6] directly modifies selected model parameters to rewrite factual associations, and task arithmetic [7] steers pretrained-model behavior through parameter-space task vectors. These approaches demonstrate the broader feasibility of targeted behavioral modification, but they address different editing mechanisms and problem formulations from the EWC+KD repair procedure studied here. In particular, our focus is not on editing factual associations in language models or learning a separate editor; it is on repairing a behavioral failure in an existing policy using a small corrective set plus limited anchors while explicitly accounting for anchor contamination.

The closest methodological precedent is [4], who combine a task-learning loss with EWC and KD to balance new-task learning against retention in a continual-classification setting. Our problem differs in framing: rather than sequential task learning, we address targeted correction of a single identified behavioral failure under incomplete access to the original training data, and we treat the possibility that the anchor set itself contains the failure ($D_{\text{anc}} \cap \mathcal{F} \neq \varnothing$) as a first-class part of the problem rather than an incidental detail. Recent analysis of model editing also shows that repeated edits can induce forgetting and downstream performance degradation [8], reinforcing the importance of evaluating collateral behavioral change rather than reporting only edit success.

## 10 Scope Statement

**Explicitly not claimed:**

- That $D_{\text{fix}}$ or $\mathcal{F}$ admits a simple task-independent mathematical definition; failure is identified operationally from unmasked inference behavior, with task-specific or human judgment where needed.
- That preservation on $D_{\text{anc}}$ guarantees global behavioral equivalence $f_{\theta^\star}(x) \approx f_{\theta_0}(x)$ for all $x \in \mathcal{X}$; it is an empirical, regularized approximation over representative samples.
- That a specific $(\lambda_1, \lambda_2)$ setting is universally optimal; the trade-off is a function of $\rho$, $|D_{\text{fix}}|$, and $|D_{\text{anc}}|$ and should be tuned per setting.
- That the local equilibrium analysis of Section 6 is a global convergence theorem for a non-convex neural network; its convergence result applies to the stated local surrogate and to the subspace where the preservation Hessian is positive definite.
- That the empirical diagonal Fisher estimate used here is the exact Fisher matrix in every loss construction; outside the standard likelihood setting, it is explicitly a squared-gradient sensitivity statistic.
- That the current implementation's full-batch anchor evaluation is necessary to the method; it is a design choice of the present pipeline (Section 8).
- That the sequential learning setting of [4] and the single-failure-correction setting studied here are the same problem; we adopt their combined-loss mechanism but not their task-boundary framing.
- That the base, derived, and composite metrics are computed from independent measurement runs. All three metric groups are aggregated from the same $n = 64$ paired evaluation episodes described in Section 11.2; the derived and composite groups are deterministic transformations of the base group and are not independent measurements.

# 11 Experimental Design

## 11.1 Checkpoints Under Comparison

Three checkpoints are compared. All three share the same architecture, environment, and inference-time action mask; they differ only in the parameters produced by their respective optimization procedures.

- $\theta_0$ **(P)** — **Pre-repair:** the converged policy checkpoint before any behavioral repair, produced by the Future Action Prediction / Transformer-aggregator training procedure of [11].
- $\theta_{\text{FT}}$ **(F)** — **Naive fine-tuning:** obtained by optimizing $\theta_0$ on $D_{\text{fix}}$ alone, i.e. $\lambda_1 = \lambda_2 = 0$ in Section 4. This baseline isolates the effect of the corrective signal without any explicit preservation mechanism.
- $\theta^\star$ **(B)** — **EWC+KD repair:** the checkpoint produced by the combined objective with $\lambda_1 = \lambda_2 = 5$ (Section 8.4), implemented as described in Section 8.

Throughout this section we use the short names **P**, **F**, and **B** for $\theta_0$, $\theta_{\text{FT}}$, and $\theta^\star$ respectively.

### 11.2 Evaluation Protocol

Evaluation uses a fixed set of $n = 64$ paired episodes, with the heuristic inference-time action mask (`bc_inference`) applied uniformly to all three checkpoints. The same random seed, map, and initial condition are used within each paired episode, so that differences between checkpoints within a pair are attributable to the policy rather than to environment stochasticity. All results are reported as paired comparisons with $n = 64$ and are corrected for multiple comparisons using Benjamini–Hochberg FDR control over all (3 pairs) $\times$ (14 metrics) tests jointly. The analysis is implemented in the accompanying `compare_models.py` script, which also generates all LaTeX tables and figures referenced below.

**Why a paired design and not aggregate averages.** The pre-repair policy exhibits high per-episode variance in survival time (sample standard deviation of roughly $6.7 \times 10^3$ against a mean of $1.2 \times 10^4$, i.e. a coefficient of variation above 0.5). Unpaired comparisons at this variance level cannot reliably distinguish policy differences from episode-to-episode environment randomness. Pairing on identical initial conditions removes this shared variance and allows the paired $t$-test or Wilcoxon signed-rank test to detect policy-level effects that would otherwise be masked. The change from an earlier $n = 10$ protocol to the present $n = 64$ protocol was made specifically to obtain this resolution.

### 11.3 Metrics

We use three groups of metrics. All metrics are oriented so that larger values indicate a better policy; none of the metrics is an accumulator that grows with episode length.

**Base metrics.** The five quantities extracted from each episode log are [1]

$$N_{\text{kills}} \text{ (kills)}, \quad T_{\text{alive}} \text{ (frames alive)}, \quad A_{\text{HP}} \text{ (area under HP curve)},$$

$$A_{\text{stamina}} \text{ (area under stamina curve)}, \quad A_{\text{damage}} \text{ (area under damage-per-bullet curve)}.$$

These are the raw quantities on which the derived metrics below are built. They are reported for completeness but are not used as primary evidence: $A_{\text{HP}}$, $A_{\text{stamina}}$, and $A_{\text{damage}}$ are accumulators and therefore correlate with $T_{\text{alive}}$; $N_{\text{kills}}$ and $T_{\text{alive}}$ are the volume measures on which the targeted failure is defined.

**Derived metrics.** Six normalized quantities are computed from the base metrics:

$$S_1 = \frac{N_{\text{kills}}}{T_{\text{alive}}}, \qquad S_2 = \frac{N_{\text{kills}}}{A_{\text{stamina}}}, \qquad S_3 = \frac{N_{\text{kills}}^2}{T_{\text{alive}}\, A_{\text{damage}}},$$

$$r_{\text{HP}} = \frac{A_{\text{HP}}}{T_{\text{alive}}}, \qquad r_{\text{stam}} = \frac{A_{\text{stamina}}}{T_{\text{alive}}}, \qquad r_{\text{dmg}} = \frac{A_{\text{damage}}}{T_{\text{alive}}}.$$

$S_1$ is the kill rate, $S_2$ is kills per unit of accumulated stamina (an engagement-efficiency measure), and $S_3$ is a damage-normalized kill rate. $r_{\text{HP}}$, $r_{\text{stam}}$, and $r_{\text{dmg}}$ are the per-frame HP, stamina, and damage rates; larger values indicate a policy that maintains a higher combat capacity per unit time. All six are higher-is-better.

[1] hhttps://github.com/bistoyek21-ric/StrikeForce/blob/main/StrikeForce-client/StrikeForce-Lab-Protocol.md

**Composite metrics.** Three kill-weighted resource metrics are defined as

$$M_{\mathrm{HP}} = \frac{N_{\mathrm{kills}}\, A_{\mathrm{HP}}}{T_{\mathrm{alive}}}, \qquad M_{\mathrm{stam}} = \frac{N_{\mathrm{kills}}\, A_{\mathrm{stamina}}}{T_{\mathrm{alive}}}, \qquad M_{\mathrm{dmg}} = \frac{N_{\mathrm{kills}}\, A_{\mathrm{damage}}}{T_{\mathrm{alive}}}.$$

Each is the product of the raw kill count and the corresponding per-frame resource rate. This construction has two purposes. First, it makes each metric an interaction term rather than a pure resource-hoarding measure: a policy that accumulates HP but never converts it into kills receives $M_{\mathrm{HP}} \approx 0$. Second, it prevents the metric-hacking failure mode in which a policy survives for a long time by avoiding engagement; such a policy has high $A_{\mathrm{HP}}/T_{\mathrm{alive}}$ but low $N_{\mathrm{kills}}$, so its $M_{\mathrm{HP}}$ remains low. All three composite metrics are therefore anti-loafing by construction, and the composite group as a whole is the primary evidence that the EWC+KD repair produces a policy that fights rather than merely survives.

**On the choice not to report an argmax-based failure rate.** An earlier version of this study used a per-frame failure rate defined as the fraction of decisions on which the raw argmax of $f_\theta(s)$ did not match an operationally defined task-correct action. We no longer report this metric, for two reasons. First, it is not a genuine measurement of policy behavior: the argmax is a single point in the simplex, and small logit perturbations near the decision boundary can flip a decision without producing any behavioral change, so the metric can move by $\epsilon$-perturbation rather than by genuine repair. Second, the metric is self-referential: it is aggregated over the state distribution $d_\pi(s)$ induced by the policy itself, so a policy can improve its apparent failure rate by shifting $d_\pi$ toward low-risk states without changing behavior on $\mathcal{F}$. Both issues are absent from the rate-based and composite metrics above, which are aggregated over a fixed episode budget rather than over a policy-dependent trajectory length. The operational definition of failure on which $D_{\mathrm{fix}}$ was constructed remains in Section 8.3 for provenance, but the empirical evaluation of the repair is now based on the metrics below.

## 11.4 Results

Tables 1–3 report the three metric groups for P, B, and F over $n = 64$ paired episodes.

| Metric | P ($\theta_0$) | B ($\theta^\star$) | F ($\theta_{\mathrm{FT}}$) |
|---|---|---|---|
| $T_{\mathrm{alive}}$ | $11908 \pm 6714$ | $10991 \pm 5920$ | $9630 \pm 4554$ |
| $N_{\mathrm{kills}}$ | $184.1 \pm 100.0$ | $188.5 \pm 88.9$ | $167.8 \pm 78.5$ |
| $A_{\mathrm{HP}}$ | $(1.03 \pm 0.68) \times 10^8$ | $(0.99 \pm 0.59) \times 10^8$ | $(0.80 \pm 0.50) \times 10^8$ |
| $A_{\mathrm{stamina}}$ | $(1.21 \pm 0.46) \times 10^8$ | $(1.15 \pm 0.41) \times 10^8$ | $(1.05 \pm 0.35) \times 10^8$ |
| $A_{\mathrm{damage}}$ | $(2.28 \pm 1.74) \times 10^7$ | $(2.37 \pm 1.57) \times 10^7$ | $(2.05 \pm 1.34) \times 10^7$ |

Table 1: Base metrics, mean $\pm$ SD over $n = 64$ paired episodes.

| Metric | P ($\theta_0$) | B ($\theta^\star$) | F ($\theta_{\mathrm{FT}}$) |
|---|---|---|---|
| $S_1 = N_{\mathrm{kills}}/T_{\mathrm{alive}}$ | $1.566 \times 10^{-2}$ | $1.733 \times 10^{-2}$ | $\mathbf{1.762 \times 10^{-2}}$ |
| $S_2 = N_{\mathrm{kills}}/A_{\mathrm{stamina}}$ | $1.451 \times 10^{-6}$ | $\mathbf{1.587 \times 10^{-6}}$ | $1.558 \times 10^{-6}$ |
| $S_3 = N_{\mathrm{kills}}^2/(T_{\mathrm{alive}} A_{\mathrm{damage}})$ | $1.467 \times 10^{-7}$ | $1.534 \times 10^{-7}$ | $\mathbf{1.582 \times 10^{-7}}$ |
| $r_{\mathrm{HP}} = A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | 8540 | **8876** | 8243 |
| $r_{\mathrm{stam}} = A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | 11022 | 11089 | **11452** |
| $r_{\mathrm{dmg}} = A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | 1801 | **2075** | 2051 |

Table 2: Derived metrics, means over $n = 64$ paired episodes. Bold marks the best checkpoint per row.

| Metric | P ($\theta_0$) | B ($\theta^\star$) | F ($\theta_{\mathrm{FT}}$) |
|---|---|---|---|
| $M_{\mathrm{HP}} = N_{\mathrm{kills}} A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $1.596 \times 10^6$ | $\mathbf{1.690 \times 10^6}$ | $1.400 \times 10^6$ |
| $M_{\mathrm{stam}} = N_{\mathrm{kills}} A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | $1.890 \times 10^6$ | $\mathbf{1.989 \times 10^6}$ | $1.835 \times 10^6$ |
| $M_{\mathrm{dmg}} = N_{\mathrm{kills}} A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $3.577 \times 10^5$ | $\mathbf{4.095 \times 10^5}$ | $3.611 \times 10^5$ |

Table 3: Composite anti-loafing metrics, means over $n = 64$ paired episodes. Bold marks the best checkpoint per row.

### 11.5 Paired Statistical Analysis

Table 4 reports the pairwise comparisons that form the primary evidence of the study. For each pair $(M_1, M_2)$ we report the mean paired difference $\Delta = M_2 - M_1$, the FDR-adjusted $p$-value $q_{\mathrm{FDR}}$, and Cohen's $d_z$. The full per-metric comparisons for each of the three pairs are provided in the appendix (Tables 11.7–11.7).

| Pair | Metric | $\Delta$ | $q_{\mathrm{FDR}}$ | $d_z$ |
|---|---|---|---|---|
| P vs B | $S_1 = N_{\mathrm{kills}}/T_{\mathrm{alive}}$ | $+1.67 \times 10^{-3}$ | $< 0.001$ | $+0.62$ |
| P vs B | $S_2 = N_{\mathrm{kills}}/A_{\mathrm{stamina}}$ | $+1.36 \times 10^{-7}$ | 0.011 | $+0.44$ |
| P vs B | $r_{\mathrm{dmg}} = A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $+274$ | 0.023 | $+0.40$ |
| P vs B | $T_{\mathrm{alive}}$ | $-917$ | 0.591 | $-0.11$ |
| P vs B | $r_{\mathrm{HP}} = A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $+336$ | 0.394 | $+0.16$ |
| P vs B | $M_{\mathrm{HP}}$ | $+9.35 \times 10^4$ | 0.686 | $+0.08$ |
| P vs F | $S_1 = N_{\mathrm{kills}}/T_{\mathrm{alive}}$ | $+1.96 \times 10^{-3}$ | $< 0.001$ | $+0.67$ |
| P vs F | $r_{\mathrm{dmg}} = A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $+250$ | 0.053 | $+0.34$ |
| P vs F | $r_{\mathrm{HP}} = A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $-297$ | 0.504 | $-0.12$ |
| P vs F | $M_{\mathrm{HP}}$ | $-1.96 \times 10^5$ | 0.394 | $-0.13$ |
| B vs F | $r_{\mathrm{HP}} = A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $-633$ | 0.066 | $-0.30$ |
| B vs F | $A_{\mathrm{HP}}$ | $-1.85 \times 10^7$ | 0.029 | $-0.24$ |
| B vs F | $M_{\mathrm{HP}}$ | $-2.90 \times 10^5$ | 0.023 | $-0.22$ |
| B vs F | $S_1 = N_{\mathrm{kills}}/T_{\mathrm{alive}}$ | $+2.85 \times 10^{-4}$ | 0.686 | $+0.08$ |

Table 4: Selected paired comparisons. $\Delta = M_2 - M_1$; positive means $M_2$ is larger. $q_{\mathrm{FDR}}$ is the Benjamini–Hochberg adjusted $p$-value.

### 11.6 Figures

Three figures summarize the comparison. The radar chart (Figure 1) is the primary visual summary; the Pareto frontier (Figure 2) shows the dominance structure; and the forest plots (Figure 3) give

the per-metric effect sizes.

**Radar summary (Figure 1).** Figure 1 is the compact visual summary of the comparison and the graphical equivalent of a ten-dimension scorecard. Each dimension is min-max normalized across the three checkpoints and oriented so that "outward" always means "better." B is the outermost curve on seven of the ten axes, F on three, and P on none; the axis-count ordering reported in the interpretation below is read directly from this figure. The radar chart replaces a separate scorecard table: the same win/loss information is conveyed more compactly, and the shape of each polygon shows which dimensions drive each checkpoint's position.

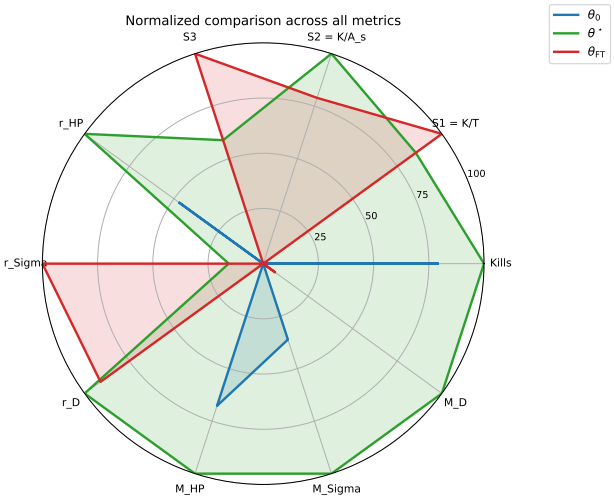



Figure 1: Direction-aware radar summary across the ten scorecard dimensions ($N_{\text{kills}}$, $S_1$, $S_2$, $S_3$, $r_{\text{HP}}$, $r_{\text{stam}}$, $r_{\text{dmg}}$, $M_{\text{HP}}$, $M_{\text{stam}}$, $M_{\text{dmg}}$). Each axis is min-max normalized across the three checkpoints and oriented so that outward is better. B occupies the largest area, followed by F, with P innermost.

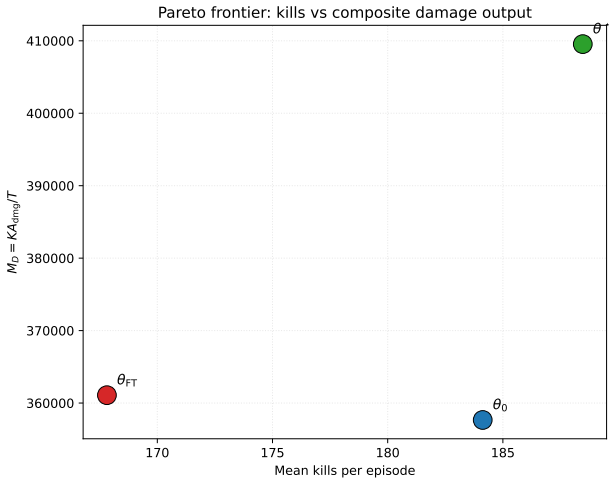



Figure 2: Pareto frontier in the ($N_{\text{kills}}$, composite damage output $M_{\text{dmg}}$) plane. Only B is non-dominated; both P and F are dominated by B on this two-dimensional objective.

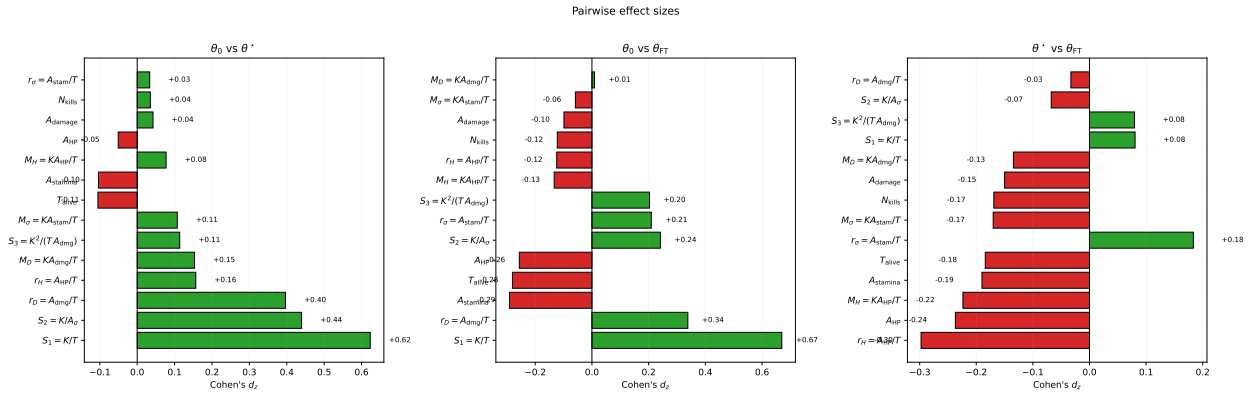



Figure 3: Pairwise effect sizes ($d_z$) for all metrics and all three pairs. Left: P vs B. Center: P vs F. Right: B vs F. Bars are oriented left for negative effects and right for positive.

**Pareto frontier (Figure 2).** Figure 2 plots each checkpoint in the two-dimensional space (mean $N_{\text{kills}}$ per episode, composite damage output $M_{\text{dmg}}$). Only B is Pareto-optimal: P is dominated by B (B has both higher $N_{\text{kills}}$ and higher $M_{\text{dmg}}$), and F is dominated by B as well. This formalizes the sense in which the EWC+KD repair is the unique non-dominated checkpoint in the comparison.

**Forest plots (Figure 3).** Figure 3 shows side-by-side forest plots of the pairwise effect sizes for all metrics and all three pairs. The leftmost panel (P vs B) shows a systematic shift to the right, with only $T_{\text{alive}}$ and $A_{\text{stamina}}$ slightly negative; the middle panel (P vs F) shows a mixed pattern with several negative preservation metrics; and the rightmost panel (B vs F) shows a clean separation in which the significant effects are all in the preservation direction.

### 11.7 Interpretation

**B strictly dominates P on the targeted axis and preserves everything else.** Against the pre-repair checkpoint P, the EWC+KD repair B improves on all ten dimensions of the direction-aware radar summary (Figure 1), with three of the ten reaching FDR-corrected significance: kill rate $S_1$ (+10.7%, $d_z = 0.62$, medium), kill per stamina $S_2$ (+9.3%, $d_z = 0.44$, small), and damage rate $r_{\text{dmg}}$ (+15.2%, $d_z = 0.40$, small). Among the three checkpoints, B is best on seven of the ten axes, F on three, and P on none. The preservation metrics $T_{\text{alive}}$, $r_{\text{HP}}$, $r_{\text{stam}}$, and the composites $M_{\text{HP}}$, $M_{\text{stam}}$, $M_{\text{dmg}}$ do not change significantly ($q > 0.39$ for all six), which is the intended behavior of the preservation terms: the repair moves the model in the corrective direction, and the preservation geometry suppresses movement in directions that would degrade general combat capacity.

**Naive fine-tuning achieves the same engagement gain at a measurable preservation cost.** F also increases $S_1$ over P (+12.5%, $d_z = 0.67$), and its $S_1$ is not significantly different from B's ($q = 0.69$, $d_z = 0.08$). The two repaired checkpoints are therefore statistically indistinguishable on the targeted engagement axis. The distinction between them lies in preservation. On the preservation metrics, B beats F: total HP $A_{\text{HP}}$ ($q = 0.029$, $d_z = -0.24$) and the composite $M_{\text{HP}}$ ($q = 0.023$, $d_z = -0.22$) are both significantly larger under B, and the HP rate $r_{\text{HP}}$ is marginally so ($q = 0.066$, $d_z = -0.30$). Naive fine-tuning thus pays for its engagement gain with a measurable reduction in the HP capacity that the pre-repair checkpoint retained. This is the trade-off the local theory is intended to explain: removing $H_R$ allows the corrective gradient to displace the model further,

and the additional displacement along directions the preservation terms would otherwise suppress shows up as degraded preservation rather than as additional engagement gain. The engagement improvement itself is the definition of the repair task, not a prediction of the theory; what the theory accounts for is why the same engagement gain is obtained at a measurable preservation cost when $H_R$ is removed.

**Pareto frontier.** Considering the two-dimensional objective (raw $N_{\text{kills}}$, composite damage output $M_{\text{dmg}}$), only B is Pareto-optimal. P is dominated by B (B has both higher $N_{\text{kills}}$ and higher $M_{\text{dmg}}$), and F is dominated by B (B has higher $N_{\text{kills}}$ and higher $M_{\text{dmg}}$). The relative composite score, defined as the mean of the three composite metrics normalized by their maximum, summarizes this ordering: $P = 0.923$, $B = 1.000$, $F = 0.878$.

**Consistency with the local theory.** The empirical pattern is consistent with the local picture of Section 6: the repair does not follow the corrective gradient indefinitely, but settles at an equilibrium in which the corrective force is balanced against the total curvature. The theory is used here as an explanation of the preservation result, not as a prediction of the targeted-axis improvement, which follows from the construction of $D_{\text{fix}}$. In this equilibrium, the components of the corrective gradient along low-curvature directions are realized (engagement improves), while components along high-curvature directions are suppressed (survival, HP retention, and stamina retention are unchanged). Under this reading, the comparison between B and F is a direct illustration of the role of $H_R$ in the local surrogate $\Delta^\star = -H_{\text{total}}^{-1} g_F$.

**What the results do not claim.** The aggregate difference between B and F on engagement is not significant at $n = 64$ ($d_z = 0.08$ for $S_1$). We therefore do not claim that B strictly outperforms F on engagement; the two are statistically equivalent on the targeted axis, and B is strictly better on the preservation axis. The value of the EWC+KD repair is not a larger correction but a correction of the same magnitude at a lower preservation cost. This is the sense in which the repair is *targeted*: it moves the policy on the intended behavioral dimension while leaving the rest of the behavior distribution statistically intact.

Paired comparison: M1 vs M2, $\Delta = M2 - M1$. $n = 64$. wins = (M1 wins):(M2 wins).

| Metric | $\Delta = \theta^\star - \theta_0$ | $q_{\mathrm{FDR}}$ | $d_z$ | size | wins |
|---|---|---|---|---|---|
| $S_1 = N_{\mathrm{kills}}/T_{\mathrm{alive}}$ | $+0.00167182$ | $< 0.001$ | 0.62 | medium | 17:47 |
| $S_2 = N_{\mathrm{kills}}/A_{\mathrm{stamina}}$ | $+1.35721e-07$ | 0.0114 | 0.44 | small | 21:43 |
| $r_{\mathrm{dmg}} = A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $+274.319$ | 0.0225 | 0.40 | small | 27:37 |
| $r_{\mathrm{HP}} = A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $+335.846$ | 0.3937 | 0.16 | negligible | 29:35 |
| $M_{\mathrm{dmg}} = N_{\mathrm{kills}}A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $+51879.3$ | 0.3937 | 0.15 | negligible | 28:36 |
| $S_3 = N_{\mathrm{kills}}^2/(T_{\mathrm{alive}}A_{\mathrm{damage}})$ | $+6.74418e-09$ | 0.5912 | 0.11 | negligible | 24:40 |
| $M_{\mathrm{stam}} = N_{\mathrm{kills}}A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | $+98306.7$ | 0.5912 | 0.11 | negligible | 32:32 |
| $T_{\mathrm{alive}}$ | $-916.844$ | 0.5912 | $-0.11$ | negligible | 37:27 |
| $A_{\mathrm{stamina}}$ | $-6.40872e+06$ | 0.5912 | $-0.10$ | negligible | 37:27 |
| $M_{\mathrm{HP}} = N_{\mathrm{kills}}A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $+93508.9$ | 0.6857 | 0.08 | negligible | 31:33 |
| $A_{\mathrm{HP}}$ | $-4.29082e+06$ | 0.7785 | $-0.05$ | negligible | 35:29 |
| $A_{\mathrm{damage}}$ | $+966423$ | 0.7935 | 0.04 | negligible | 33:31 |
| $N_{\mathrm{kills}}$ | $+4.34375$ | 0.7935 | 0.04 | negligible | 34:29 |
| $r_{\mathrm{stam}} = A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | $+67.1187$ | 0.7935 | 0.03 | negligible | 35:29 |

| Metric | $\Delta = \theta_{\mathrm{FT}} - \theta_0$ | $q_{\mathrm{FDR}}$ | $d_z$ | size | wins |
|---|---|---|---|---|---|
| $S_1 = N_{\mathrm{kills}}/T_{\mathrm{alive}}$ | $+0.00195713$ | $< 0.001$ | 0.67 | medium | 20:44 |
| $r_{\mathrm{dmg}} = A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $+249.833$ | 0.0525 | 0.34 | small | 27:37 |
| $A_{\mathrm{stamina}}$ | $-1.6373e+07$ | 0.1053 | $-0.29$ | small | 38:26 |
| $T_{\mathrm{alive}}$ | $-2277.75$ | 0.2307 | $-0.28$ | small | 36:28 |
| $A_{\mathrm{HP}}$ | $-2.27595e+07$ | 0.1215 | $-0.26$ | small | 37:27 |
| $S_2 = N_{\mathrm{kills}}/A_{\mathrm{stamina}}$ | $+1.07132e-07$ | 0.1053 | 0.24 | small | 24:40 |
| $r_{\mathrm{stam}} = A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | $+429.849$ | 0.2447 | 0.21 | small | 28:36 |
| $S_3 = N_{\mathrm{kills}}^2/(T_{\mathrm{alive}}A_{\mathrm{damage}})$ | $+1.14717e-08$ | 0.2561 | 0.20 | small | 26:38 |
| $M_{\mathrm{HP}} = N_{\mathrm{kills}}A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $-196167$ | 0.3937 | $-0.13$ | negligible | 35:29 |
| $r_{\mathrm{HP}} = A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $-297.025$ | 0.5039 | $-0.12$ | negligible | 34:30 |
| $N_{\mathrm{kills}}$ | $-16.3125$ | 0.6926 | $-0.12$ | negligible | 33:31 |
| $A_{\mathrm{damage}}$ | $-2.26949e+06$ | 0.7223 | $-0.10$ | negligible | 32:32 |
| $M_{\mathrm{stam}} = N_{\mathrm{kills}}A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | $-55673.2$ | 0.7498 | $-0.06$ | negligible | 31:33 |
| $M_{\mathrm{dmg}} = N_{\mathrm{kills}}A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $+3421.67$ | 0.7935 | 0.01 | negligible | 32:32 |

| Metric | $\Delta = \theta_{\mathrm{FT}} - \theta^\star$ | $q_{\mathrm{FDR}}$ | $d_z$ | size | wins |
|---|---|---|---|---|---|
| $r_{\mathrm{HP}} = A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $-632.872$ | 0.0662 | $-0.30$ | small | 41:23 |
| $A_{\mathrm{HP}}$ | $-1.84687e+07$ | 0.0294 | $-0.24$ | small | 43:21 |
| $M_{\mathrm{HP}} = N_{\mathrm{kills}}A_{\mathrm{HP}}/T_{\mathrm{alive}}$ | $-289676$ | 0.0225 | $-0.22$ | small | 45:19 |
| $A_{\mathrm{stamina}}$ | $-9.96425e+06$ | 0.2953 | $-0.19$ | negligible | 39:25 |
| $T_{\mathrm{alive}}$ | $-1360.91$ | 0.2447 | $-0.18$ | negligible | 39:25 |
| $r_{\mathrm{stam}} = A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | $+362.73$ | 0.3089 | 0.18 | negligible | 26:38 |
| $M_{\mathrm{stam}} = N_{\mathrm{kills}}A_{\mathrm{stamina}}/T_{\mathrm{alive}}$ | $-153980$ | 0.3567 | $-0.17$ | negligible | 37:27 |
| $N_{\mathrm{kills}}$ | $-20.6562$ | 0.2447 | $-0.17$ | negligible | 36:28 |
| $A_{\mathrm{damage}}$ | $-3.23591e+06$ | 0.2447 | $-0.15$ | negligible | 39:25 |
| $M_{\mathrm{dmg}} = N_{\mathrm{kills}}A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $-48457.6$ | 0.2447 | $-0.13$ | negligible | 40:24 |
| $S_1 = N_{\mathrm{kills}}/T_{\mathrm{alive}}$ | $+0.000285309$ | 0.6857 | 0.08 | negligible | 32:32 |
| $S_3 = N_{\mathrm{kills}}^2/(T_{\mathrm{alive}}A_{\mathrm{damage}})$ | $+4.72752e-09$ | 0.6857 | 0.08 | negligible | 30:34 |
| $S_2 = N_{\mathrm{kills}}/A_{\mathrm{stamina}}$ | $-2.85897e-08$ | 0.5912 | $-0.07$ | negligible | 35:29 |
| $r_{\mathrm{dmg}} = A_{\mathrm{damage}}/T_{\mathrm{alive}}$ | $-24.4861$ | 0.7935 | $-0.03$ | negligible | 34:30 |

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